In [2]:
```
import pandas as pd
```

In [3]:
```
!pip install openpyxl
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: openpyxl in c:\programdata\anaconda3\lib\site-packag
es (3.1.5)
Requirement already satisfied: et-xmlfile in c:\programdata\anaconda3\lib\site-pack
ages (from openpyxl) (2.0.0)
```

In [4]:
```
import pandas as pd
import os

# Ye tere Downloads folder ki saari Excel files dikha dega
downloads_path = os.path.expanduser("~/Downloads")
print("Downloads me ye files hain:")
for file in os.listdir(downloads_path):
    if "SQL" in file:
        print(file)
```

```
Downloads me ye files hain:
Copy of SQL_Sales_Dataset_200_Rows.csv
Copy of SQL_Sales_Dataset_200_Rows.xlsx
MySQL-Logo.wine.svg
```

In [5]:
```
import os

file_path = os.path.join(os.path.expanduser("~/Downloads"), "Copy of SQL_Sales_Dat

df = pd.read_excel(file_path)

df.head()
```

Out[5]:

	order_id	customer_name	order_date	category	sub_category	product_name	quanti
0	1000	Caleb Davis	2025-01-31	Clothing	Jeans	Jeans Try	
1	1001	Misty Herrera	2025-05-05	Clothing	Jeans	Jeans New	
2	1002	Rick Soto	2025-06-09	Clothing	Shoes	Shoes Us	
3	1003	Sean Haas	2025-02-27	Furniture	Desk	Desk Section	
4	1004	Ann Hall	2025-06-01	Furniture	Cabinet	Cabinet Center	

In [6]:
```
df.shape
```

Out[6]:
```
(200, 10)
```

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype         
---  ------         --------------  -----         
 0   order_id       200 non-null    int64         
 1   customer_name  200 non-null    object        
 2   order_date     200 non-null    datetime64[ns]
 3   category       200 non-null    object        
 4   sub_category   200 non-null    object        
 5   product_name   200 non-null    object        
 6   quantity       200 non-null    int64         
 7   unit_price     200 non-null    int64         
 8   total_price    200 non-null    int64         
 9   region         200 non-null    object        
dtypes: datetime64[ns](1), int64(4), object(5)
memory usage: 15.8+ KB
```

In [8]:
```
print(df.isnull().sum())
```

```
order_id         0
customer_name    0
order_date       0
category         0
sub_category     0
product_name     0
quantity         0
unit_price       0
total_price      0
region           0
dtype: int64
```

In [9]:
```
print(df.columns.tolist())
```

```
['order_id', 'customer_name', 'order_date', 'category', 'sub_category', 'product_na
me', 'quantity', 'unit_price', 'total_price', 'region']
```

In [10]:
```
# missing value
df['unit_price'] = df['unit_price'].fillna(df['unit_price'].mean())
df['total_price'] = df['total_price'].fillna(df['total_price'].mean())
df['quantity'] = df['quantity'].fillna(df['quantity'].mean())
```

In [11]:
```
df
```

Out[11]:

	order_id	customer_name	order_date	category	sub_category	product_name	qua
0	1000	Caleb Davis	2025-01-31	Clothing	Jeans	Jeans Try	
1	1001	Misty Herrera	2025-05-05	Clothing	Jeans	Jeans New	
2	1002	Rick Soto	2025-06-09	Clothing	Shoes	Shoes Us	
3	1003	Sean Haas	2025-02-27	Furniture	Desk	Desk Section	
4	1004	Ann Hall	2025-06-01	Furniture	Cabinet	Cabinet Center	
...	...	...	...	...	...	...	
195	1195	Dylan Baxter	2025-02-15	Clothing	Jeans	Jeans Sound	
196	1196	Brandon Callahan	2025-02-26	Grocery	Snacks	Snacks Lead	
197	1197	Angela Stanton	2025-06-16	Furniture	Cabinet	Cabinet None	
198	1198	Marissa Aguilar	2025-06-02	Grocery	Milk	Milk Listen	
199	1199	Timothy Wheeler	2025-01-29	Clothing	Shirt	Shirt Area	

200 rows × 10 columns

In [12]:
```
print(df.isnull().sum())
df = df.drop_duplicates()
df['unit_price'] = df['unit_price'].fillna(df['unit_price'].mean())
df['total_price'] = df['total_price'].fillna(df['total_price'].mean())
```

```
order_id        0
customer_name   0
order_date      0
category        0
sub_category    0
product_name    0
quantity        0
unit_price      0
total_price     0
region          0
dtype: int64
```

In [13]:
```
# Group data by category and find total revenue.
df.groupby('category')['total_price'].sum()
```

Out[13]:
```
category
Clothing       484259
Electronics    549302
Furniture      714399
Grocery        672147
Name: total_price, dtype: int64
```

In [14]:
```
# Sort data by multiple columns using Python.
sorted_df = df.sort_values(by=['category', 'total_price'], ascending=[True, False]
sorted_df.head()
```

Out[14]:

	order_id	customer_name	order_date	category	sub_category	product_name	qua
168	1168	Megan Charles	2025-03-26	Clothing	Jeans	Jeans Card	
59	1059	Miss Rebecca Levine	2025-02-28	Clothing	Shirt	Shirt Look	
157	1157	Patricia Gill	2025-03-23	Clothing	Jacket	Jacket Song	
180	1180	Hailey Cobb	2025-02-19	Clothing	Jacket	Jacket Heart	
82	1082	Kenneth Lee	2025-05-06	Clothing	Jacket	Jacket Realize	

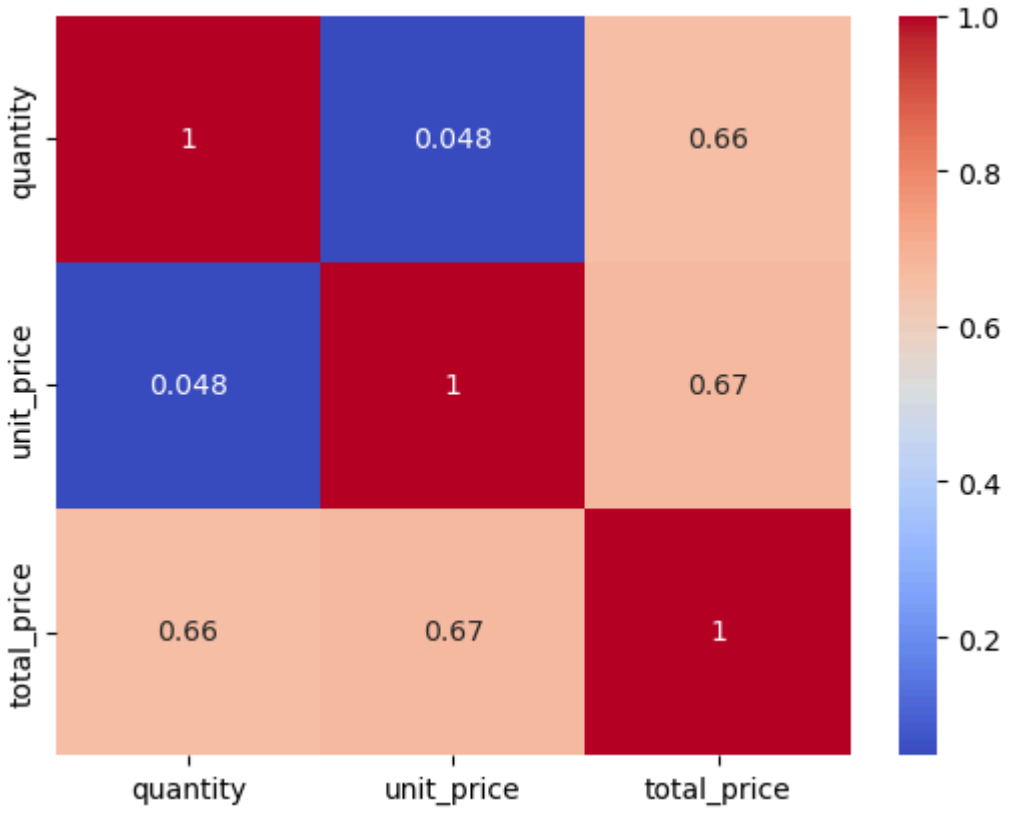
In [15]:
```
#Create a correlation matrix for numerical columns
```

In [16]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# numeric columns -> quantity, unit_price, total_price
corr = df[['quantity', 'unit_price', 'total_price']].corr()
print(corr)

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.show()
```

```
              quantity  unit_price  total_price
quantity      1.000000    0.047996     0.655212
unit_price    0.047996    1.000000     0.671318
total_price   0.655212    0.671318     1.000000
```


quantity
unit_price
total_price
1
0.048
0.66
0.048
1
0.67
0.66
0.67
1
quantity
unit_price
total_price
1.0
0.8
0.6
0.4
0.2

In [37]:
```
df
```

Out[37]:

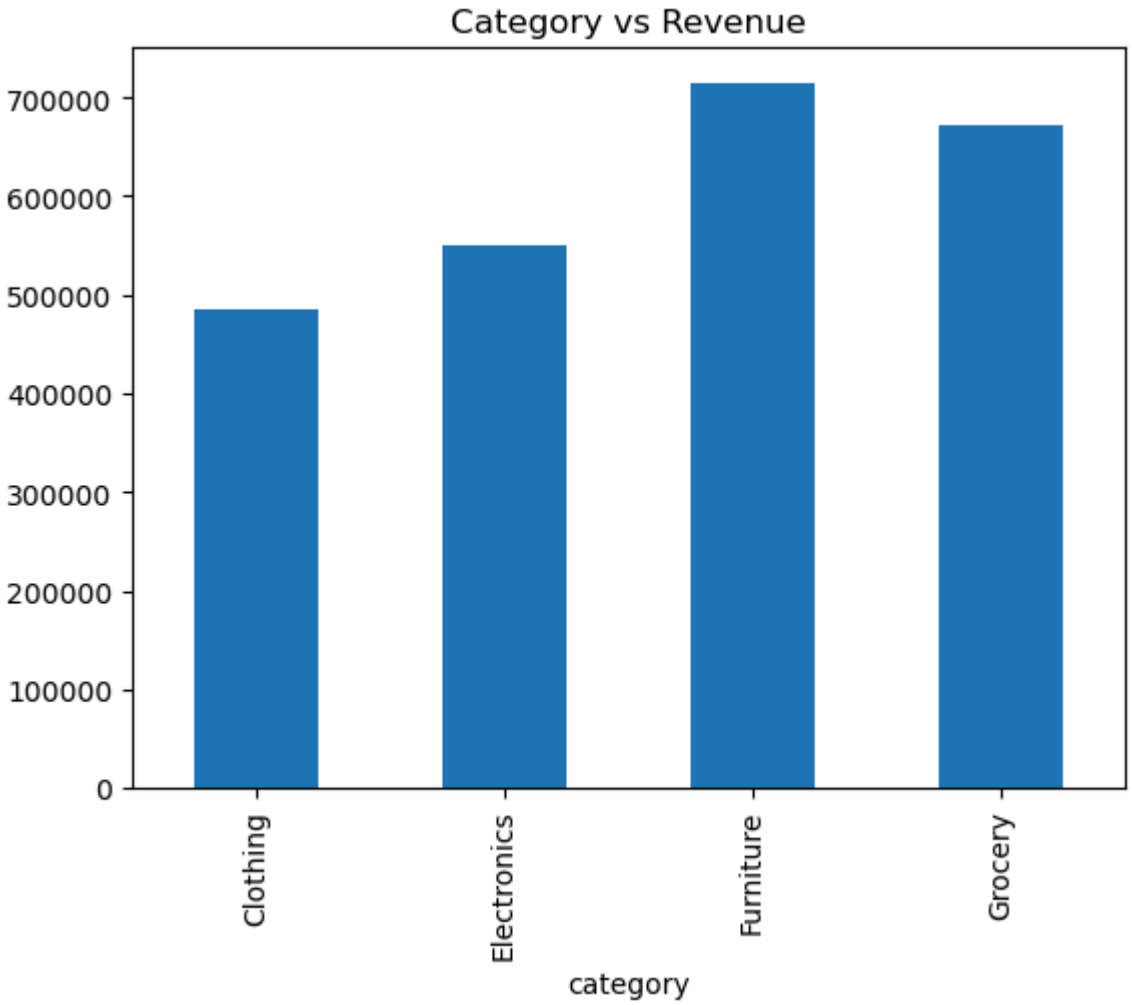
	order_id	customer_name	order_date	category	sub_category	product_name	qua
0	1000	Caleb Davis	2025-01-31	Clothing	Jeans	Jeans Try	
1	1001	Misty Herrera	2025-05-05	Clothing	Jeans	Jeans New	
2	1002	Rick Soto	2025-06-09	Clothing	Shoes	Shoes Us	
3	1003	Sean Haas	2025-02-27	Furniture	Desk	Desk Section	
4	1004	Ann Hall	2025-06-01	Furniture	Cabinet	Cabinet Center	
...	...	...	...	...	...	...	
195	1195	Dylan Baxter	2025-02-15	Clothing	Jeans	Jeans Sound	
196	1196	Brandon Callahan	2025-02-26	Grocery	Snacks	Snacks Lead	
197	1197	Angela Stanton	2025-06-16	Furniture	Cabinet	Cabinet None	
198	1198	Marissa Aguilar	2025-06-02	Grocery	Milk	Milk Listen	
199	1199	Timothy Wheeler	2025-01-29	Clothing	Shirt	Shirt Area	

200 rows × 10 columns

In [17]:
```
high_value_orders = df[df['total_price'] > 10000]
```

In [19]:
```
df['price_after_discount'] = df['total_price'] * 0.9
```

In [20]:
```
import matplotlib.pyplot as plt
df.groupby('category')['total_price'].sum().plot(kind='bar')
plt.title('Category vs Revenue')
plt.show()
```



In [21]:
```
!pip install nbconvert
!jupyter nbconvert --to pdf Untitled.ipynb
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: nbconvert in c:\programdata\anaconda3\lib\site-packa
ges (7.16.6)
Requirement already satisfied: beautifulsoup4 in c:\programdata\anaconda3\lib\site-
packages (from nbconvert) (4.13.5)
Requirement already satisfied: bleach!=5.0.0 in c:\programdata\anaconda3\lib\site-p
ackages (from bleach[css]!=5.0.0->nbconvert) (6.3.0)
Requirement already satisfied: defusedxml in c:\programdata\anaconda3\lib\site-pack
ages (from nbconvert) (0.7.1)
Requirement already satisfied: jinja2>=3.0 in c:\programdata\anaconda3\lib\site-pac
kages (from nbconvert) (3.1.6)
Requirement already satisfied: jupyter-core>=4.7 in c:\programdata\anaconda3\lib\si
te-packages (from nbconvert) (5.8.1)
Requirement already satisfied: jupyterlab-pygments in c:\programdata\anaconda3\lib
\site-packages (from nbconvert) (0.3.0)
Requirement already satisfied: markupsafe>=2.0 in c:\programdata\anaconda3\lib\site
-packages (from nbconvert) (3.0.2)
Requirement already satisfied: mistune<4,>=2.0.3 in c:\programdata\anaconda3\lib\si
te-packages (from nbconvert) (3.1.2)
Requirement already satisfied: nbclient>=0.5.0 in c:\programdata\anaconda3\lib\site
-packages (from nbconvert) (0.10.2)
Requirement already satisfied: nbformat>=5.7 in c:\programdata\anaconda3\lib\site-p
ackages (from nbconvert) (5.10.4)
Requirement already satisfied: packaging in c:\programdata\anaconda3\lib\site-packa
ges (from nbconvert) (25.0)
Requirement already satisfied: pandocfilters>=1.4.1 in c:\programdata\anaconda3\lib
\site-packages (from nbconvert) (1.5.1)
Requirement already satisfied: pygments>=2.4.1 in c:\programdata\anaconda3\lib\site
-packages (from nbconvert) (2.19.2)
Requirement already satisfied: traitlets>=5.1 in c:\programdata\anaconda3\lib\site-
packages (from nbconvert) (5.14.3)
Requirement already satisfied: webencodings in c:\programdata\anaconda3\lib\site-pa
ckages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in c:\programdata\anaconda3\lib
\site-packages (from bleach[css]!=5.0.0->nbconvert) (1.4.0)
Requirement already satisfied: platformdirs>=2.5 in c:\programdata\anaconda3\lib\si
te-packages (from jupyter-core>=4.7->nbconvert) (4.5.0)
Requirement already satisfied: pywin32>=300 in c:\programdata\anaconda3\lib\site-pa
ckages (from jupyter-core>=4.7->nbconvert) (311)
Requirement already satisfied: jupyter-client>=6.1.12 in c:\programdata\anaconda3\l
ib\site-packages (from nbclient>=0.5.0->nbconvert) (8.6.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\l
ib\site-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (2.9.0.p
ost0)
Requirement already satisfied: pyzmq>=23.0 in c:\programdata\anaconda3\lib\site-pac
kages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (27.1.0)
Requirement already satisfied: tornado>=6.2 in c:\programdata\anaconda3\lib\site-pa
ckages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (6.5.1)
Requirement already satisfied: fastjsonschema>=2.15 in c:\programdata\anaconda3\lib
\site-packages (from nbformat>=5.7->nbconvert) (2.21.2)
Requirement already satisfied: jsonschema>=2.6 in c:\programdata\anaconda3\lib\site
-packages (from nbformat>=5.7->nbconvert) (4.25.0)
Requirement already satisfied: attrs>=22.2.0 in c:\programdata\anaconda3\lib\site-p
ackages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (25.4.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\programda
ta\anaconda3\lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (20
25.9.1)
Requirement already satisfied: referencing>=0.28.4 in c:\programdata\anaconda3\lib
\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.37.0)
Requirement already satisfied: rpds-py>=0.7.1 in c:\programdata\anaconda3\lib\site-
```

```
packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.28.0)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packag
es (from python-dateutil>=2.8.2->jupyter-client>=6.1.12->nbclient>=0.5.0->nbconver
t) (1.17.0)
Requirement already satisfied: soupsieve>1.2 in c:\programdata\anaconda3\lib\site-p
ackages (from beautifulsoup4->nbconvert) (2.5)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\programdata\anaconda3
\lib\site-packages (from beautifulsoup4->nbconvert) (4.15.0)
```

```
[NbConvertApp] Converting notebook Untitled.ipynb to pdf
[NbConvertApp] Support files will be in Untitled_files\
[NbConvertApp] Making directory .\Untitled_files
[NbConvertApp] Writing 40882 bytes to notebook.tex
[NbConvertApp] Building PDF
Traceback (most recent call last):
  File "C:\ProgramData\anaconda3\Scripts\jupyter-nbconvert-script.py", line 10, in
<module>
    sys.exit(main())
             ~~~~^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\jupyter_core\application.py", li
ne 284, in launch_instance
    super().launch_instance(argv=argv, **kwargs)
    ~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\traitlets\config\application.p
y", line 1075, in launch_instance
    app.start()
    ~~~~~~~~~^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\nbconvertapp.py", line
420, in start
    self.convert_notebooks()
    ~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\nbconvertapp.py", line
597, in convert_notebooks
    self.convert_single_notebook(notebook_filename)
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\nbconvertapp.py", line
563, in convert_single_notebook
    output, resources = self.export_single_notebook(
                        ~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        notebook_filename, resources, input_buffer=input_buffer
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\nbconvertapp.py", line
487, in export_single_notebook
    output, resources = self.exporter.from_filename(
                        ~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        notebook_filename, resources=resources
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\templateexpo
rter.py", line 390, in from_filename
    return super().from_filename(filename, resources, **kw)  # type:ignore[return-v
alue]
           ~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\exporter.p
y", line 201, in from_filename
    return self.from_file(f, resources=resources, **kw)
           ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\templateexpo
rter.py", line 396, in from_file
    return super().from_file(file_stream, resources, **kw)  # type:ignore[return-va
lue]
           ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\exporter.p
y", line 220, in from_file
    return self.from_notebook_node(
           ~~~~~~~~~~~~~~~~~~~~~~~^
```

```
        nbformat.read(file_stream, as_version=4), resources=resources, **kw
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\pdf.py", lin
e 197, in from_notebook_node
    self.run_latex(tex_file)
    ~~~~~~~~~~~~~~~^^^^^^^^^^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\pdf.py", lin
e 166, in run_latex
    return self.run_command(
           ~~~~~~~~~~~~~~~~~^
        self.latex_command, filename, self.latex_count, log_error, raise_on_failure
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\ProgramData\anaconda3\Lib\site-packages\nbconvert\exporters\pdf.py", lin
e 120, in run_command
    raise OSError(msg)
OSError: xelatex not found on PATH, if you have not installed xelatex you may need
to do so. Find further instructions at https://nbconvert.readthedocs.io/en/latest/i
nstall.html#installing-tex.
```

In [ ]: